

applications to build on each other, there must necessarily be a foundation. The primary value-add of MakerDAO is the creation of a cryptocollateralized stablecoin, pegged to USD. This means the system can run completely from within the Ethereum blockchain without relying on outside centralized institutions to back, vault, and audit the stablecoin. MakerDAO is a two-token model where a governance token MKR yields voting rights on the platform and participates in value capture. The second token is a stablecoin called DAI – a staple token in the DeFi ecosystem with which many protocols integrate, including a few we will discuss later.

DAI is generated as follows. A user can deposit ETH or other supported ERC-20 assets into a *vault*, which is a smart contract that escrows collateral and keeps track of the USD-denominated value of the collateral. The user can then mint DAI up to a certain collateralization ratio on their assets. This creates a “debt” in DAI that the vault holder must pay back. The DAI is the corresponding asset that can be used any way the vault holder wishes. For example, the user can sell the DAI for cash or leverage it into more of the collateral asset⁵ and repeat the process. Due to the volatility of ETH and most collateral types, the collateralization requirement is far in excess of 100 percent and usually in the 150–200 percent range.

The basic idea underlying the DAI mechanism is not new; it is simply a collateralized debt position. For example, a homeowner in need of some liquidity can pledge their house as collateral to a bank and receive a mortgage loan structured to include a cash takeout. The price volatility of ETH is much